Akoizo Original Practice - 2027 Rotation Form 1

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

Name: _____ Date: _____

1. (2 points) Which three broad needs define habitability for life as we know it?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Oxygen, sunlight, and soil

B. A magnetic field, rings, and moons

C. Ice, methane, and nitrogen only

D. Liquid water, energy, and key chemical elements

Answer: _____

2. (2 points) A planet moves from 1 AU to 2 AU. By what factor does the starlight flux change?

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

A. It becomes 1/8 as large

B. It becomes 1/4 as large

C. It becomes 1/2 as large

D. It doubles

Answer: _____

3. (2 points) A body orbits the Sun at 9 AU. What is its period?

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 9 years

B. 81 years

C. 3 years

D. 27 years

Answer: _____

4. (2 points) Why is Venus's surface hotter than Mercury's?

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

A. Venus has active tidal heating

B. A thick CO_2 atmosphere creates a runaway greenhouse

C. Venus is closer to the Sun

D. Venus has a strong magnetic field

Answer: _____

5. (2 points) Which moon has its own internally generated magnetic field?

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Europa

B. Titan

C. Enceladus

D. Ganymede

Answer: _____

6. (2 points) Which mission returned a sample of asteroid Bennu?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. Kepler

B. OSIRIS-REx

C. Rosetta

D. Dragonfly

Answer: _____

7. (2 points) Which method directly gives a planet's radius?

Unit: Exoplanets: detection and the named systems.
Worked-case data: An Earth-sized planet transits two stars: a Sun-like star and a red dwarf with one-tenth the Sun's radius. Compare the transit depths.
Separate investigation observations:
Light curve: Seven different periodic dips in brightness.
Star: An ultracool red dwarf, about 40 light-years away.
Sizes: All planets are roughly Earth-sized.
JWST: The two innermost planets show no thick atmosphere.

A. Radial velocity

B. Microlensing alone

C. Astrometry

D. Transit

Answer: _____

8. (2 points) Why are small, hot worlds poor at keeping light gases?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Gravity is too strong

B. Light gases freeze out

C. The Sun adds gas

D. Gas molecules move fast compared with a low escape velocity

Answer: _____

9. (2 points) If a star's surface temperature doubles, its emitted power per area changes by:

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

A. 8 times

B. 16 times

C. 2 times

D. 4 times

Answer: _____

10. (2 points) If the distance between two masses triples, the gravitational force becomes:

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

A. 1/3 as large

B. 3 times as large

C. 1/27 as large

D. 1/9 as large

Answer: _____

11. (2 points) In a simplified habitable-zone model, stellar luminosity increases by a factor of four. By what factor must distance increase to preserve incident flux?

Answer: _____
_____
_____

12. (2 points) A synthetic planet and star have radius ratio 1/100. For a uniform stellar disk and central transit, what fraction of light is blocked?

Answer: _____
_____
_____

13. (2 points) Compare surface-water and subsurface-water habitability arguments using Europa or Enceladus as listed examples.

Answer: _____
_____
_____

14. (2 points) Titan has surface liquid hydrocarbons and water ice. Explain why habitability questions should specify the solvent and biological assumptions.

Answer: _____
_____
_____

15. (2 points) A spectrum from a Mars teaching dataset shows a mineral associated with past water alteration. What can and cannot be concluded?

Answer: _____
_____
_____

16. (2 points) Explain the difference between evidence for organic molecules on Bennu or comet 67P and proof of biology.

Answer: _____
_____
_____

17. (2 points) Why can a planet in the TRAPPIST-1 system require atmospheric-retention evidence in addition to an estimated equilibrium temperature?

Answer: _____
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_____

18. (2 points) A planet is detected through periodic stellar radial-velocity shifts. What physical interaction produces the signal?

Answer: _____
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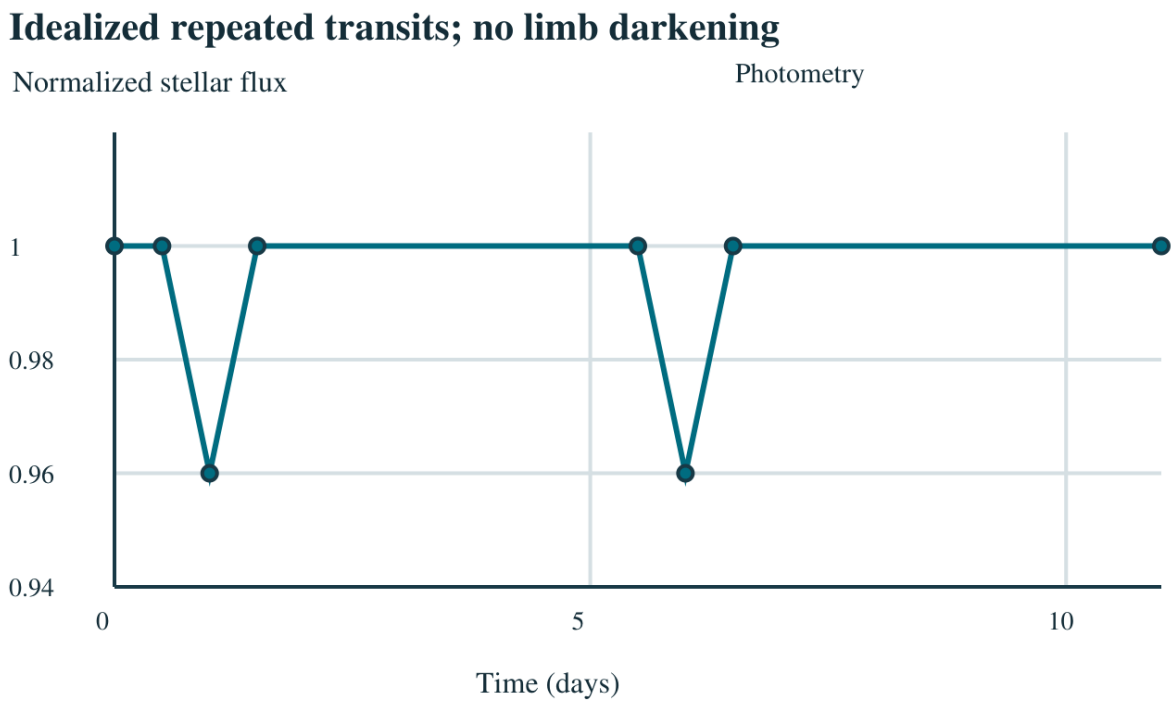
19. (2 points) Explain one engineering tradeoff between high spatial resolution and broad repeated mapping for an allowed mission.

Answer: _____
_____
_____

20. (2 points) Why is chemical disequilibrium potentially relevant to habitability but not a unique biosignature?

Answer: _____
_____
_____

21. (2 points) Use Figure F1. Read the time between the two transit minima.



F1. Exoplanet transit light curve (original synthetic teaching figure)

Answer: _____
_____
_____

22. (2 points) Use Figure F1. Find transit depth as a percentage of baseline light.

Refer to F1. Exoplanet transit light curve (original synthetic teaching figure) shown with the first question in this figure group.

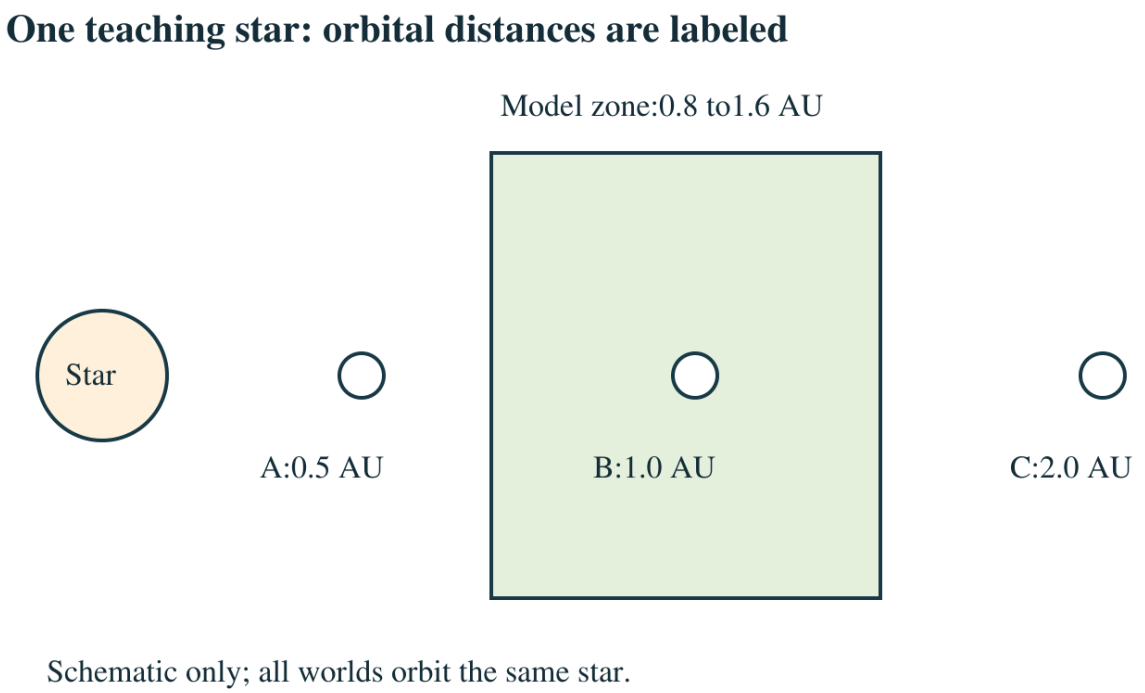
Answer: _____
_____
_____

23. (2 points) Use Figure F1. Use depth=(planet radius/star radius)^2. The stellar radius is 5 units; find planet radius. Hint:0.2 squared=0.04.

Refer to F1. Exoplanet transit light curve (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

24. (2 points) Use Figure F2. Which candidate lies inside the indicated model zone?



F2. Habitable-zone placement (original synthetic teaching figure)

Answer: _____
_____
_____

25. (2 points) Use Figure F2. Explain why that placement does not establish habitability by itself.

Refer to F2. Habitable-zone placement (original synthetic teaching figure) shown with the first question in this figure group.

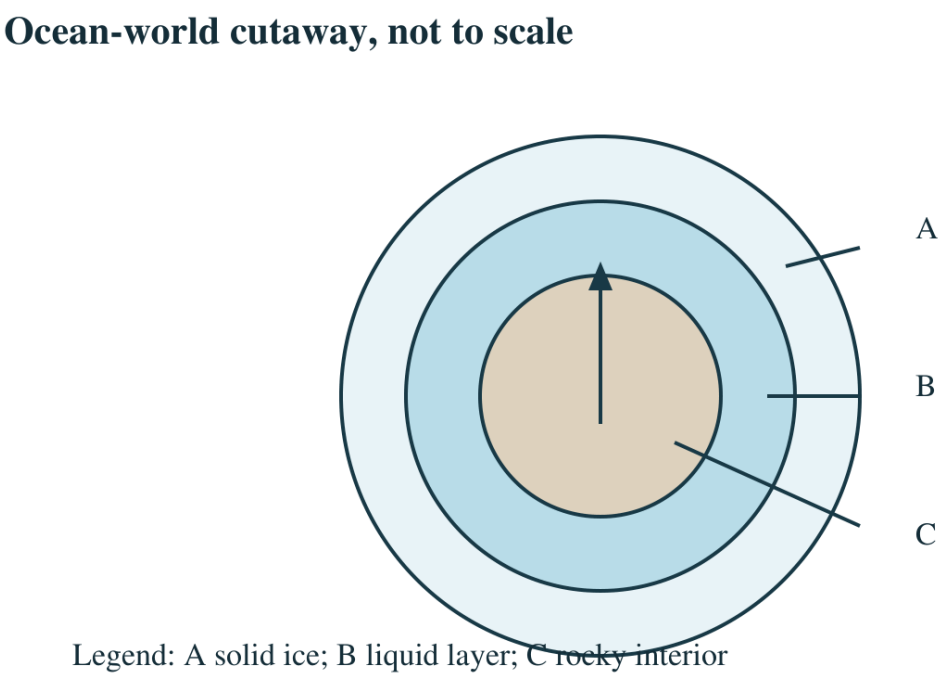
Answer: _____
_____
_____

26. (2 points) Use Figure F2. Use the labeled orbital distances to find incident stellar flux at candidate B / flux at candidate C. Use inverse-square scaling.

Refer to F2. Habitable-zone placement (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

27. (2 points) Use Figure F3. Which labeled layer is the direct liquid-water habitat candidate?



F3. Icy-world interior hypothesis (original synthetic teaching figure)

Answer: _____
_____
_____

28. (2 points) Use Figure F3. Explain one interior energy source consistent with maintaining B beneath ice.

Refer to F3. Icy-world interior hypothesis (original synthetic teaching figure) shown with the first question in this figure group.

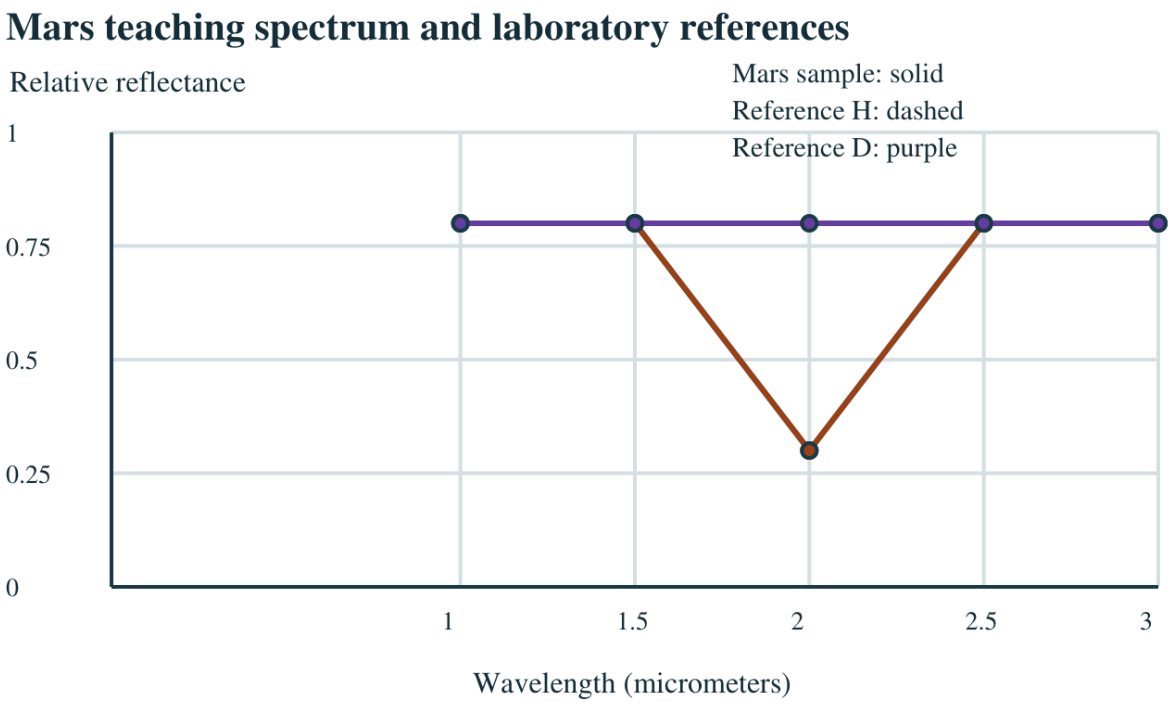
Answer: _____
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29. (2 points) Use Figure F3. Would this interior hypothesis establish living organisms? Explain.

Refer to F3. Icy-world interior hypothesis (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

30. (2 points) Use Figure F4. Which laboratory reference matches the plotted absorption?



F4. Matched mineral spectra (original synthetic teaching figure)

Answer: _____
_____
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31. (2 points) Use Figure F4. Read the sample reflectance at the absorption minimum.

Refer to F4. Matched mineral spectra (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
_____

32. (2 points) Use Figure F4. Explain why matching the hydrated-mineral reference supports water-related geological investigation without proving present life.

Refer to F4. Matched mineral spectra (original synthetic teaching figure) shown with the first question in this figure group.

Answer: _____
_____
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